

Principles of Biology 413

Fall 2016

Instructor: Dr. Kyle MacLea, kyle.maclea@unh.edu, (603) 641-4129.

Office hours: Always by appointment as well as Wednesday 3-4pm and Thursday 10am-12pm. Usually in my office (Room 559) but may occasionally move to other campus locations with notice (or a note on my office door).

Textbook: *Campbell Biology in Focus* (2014) by Urry, *et al.*, 1st edition in any version you choose (hardcover, looseleaf, ebook, etc.). You may choose to have a Mastering Biology Access Code included, because this is a very useful tool you can use to study content and prepare for exams, but it is optional. (<http://www.masteringbiology.com>). The 2nd edition of *Campbell* is also acceptable.

Lab Manual: *none*. (If you bought a lab manual by mistake, return it to the bookstore.)

Lab Coat: A lab coat is required for admittance to lab and must be purchased.

Lecture: Tuesdays 6-9 p.m in Room 301.

Preparation: As we meet only once per week it is a week's worth of material presented in one class period. To achieve success in this course it is necessary to read the material before class and review the material daily. (I know you are thinking there is no way you will read ahead of time, but because we will move through material quickly, 3 hours of material can become quickly overwhelming if you are not even vaguely familiar with it. It will also help you to come up with questions for concepts you don't understand, and have them ready to ask me in class.) You may also wish to review the chapter material on the Mastering Biology website before lecture. I highly encourage you to go to a study group weekly. Always ask questions if you do not understand!

Academic Support/Tutoring

The Center for Academic Enrichment (CAE) [in the 2nd floor Learning Commons] is a resource for all students enrolled in UNH Manchester courses. Their mission is to help under-prepared students prepare, prepared students advance, and advanced students excel. Trained peer tutors and professional staff provide free assistance with reading and writing in all courses, as well as content in all math and science courses. Support for other content areas is dependent on the background and competencies of current peer tutor staff. All enrolled students are entitled to 1 hour of free one-to-one tutoring per week, per course. In addition, peer and professional tutors may run study groups for your class and provide drop-in math tutoring. Please visit the CAE online at <http://manchester.unh.edu/student/center-academic-enrichment> to learn more about their services or to make an appointment with a tutor.

Writing: You also need to take advantage of the CAE writing tutors to turn in high quality writing, which will be assigned and turned in during lab periods. Oftentimes students do well on the exams and quizzes and poor paper writing brings down their grade—your goal is to learn scientific writing and style to prepare you for the second semester and beyond, where you will be expected to have mastered the (basic) writing concepts you learn in this course.

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MyCourses and Email: Check MyCourses (Canvas) every day. Non-urgent announcements will be posted on MyCourses. Urgent/time sensitive announcements will be sent, where possible, via email. Check your UNH email account often. This is the one I will use to contact you outside of class.

Quizzes: The timed weekly (multiple choice) quizzes are located in MyCourses and should be taken within the week after lecture. For example, since the lecture is Tuesday the quiz is typically available Tuesday at 9pm and ends the following Monday at midnight. Always take the quiz as they cannot be made up. The three lowest grades of quizzes taken will be dropped. If you do not take a quiz the grade (zero) will count. There are no makeups of quizzes.

Exams: All exams must be taken when scheduled. Make-up exams will be permitted only for excused absences (*e.g.*, confirmed illness with doctor's note) and advance notice of absence. Each exam only covers material since the previous exam; *i.e.*, the second and third exams are not cumulative. Your course grade will be determined based on the percentage of points earned out of the possible 500. 90% and above will earn A- or better; 80% and above will earn B- or better; 70% and above will earn C- or better; 60% and above will earn D- or better. This is college, so there are no extra credit assignments.

Attendance: Attendance in both lecture and laboratory is expected. If you miss one lecture, you miss 3 hours of instruction, so missing class is a huge detriment to your understanding. Although I usually do not take attendance during the lecture (*exception:* the first two weeks of class I will call the roster and verify attendance), you are responsible for all information and material presented during the class. Since the laboratory is a hands-on experience, attendance is taken. Excessive absences (two or more) from the laboratory will result in a reduction in your grade.

Disabilities: The University is committed to providing students with documented disabilities equal access to all university programs and facilities. If you think you have a disability requiring accommodations, you must register with the Disability Services Office. The Disability Services Coordinator at UNHM is Jenessa Zurek. Jenessa can be contacted at (603) 641-4170, jenessa.zurek@unh.edu, or in person in the Student Services Suite, Room 410H.

Academic Honesty: The Biology Program at UNH Manchester will strictly adhere to the University policy on academic honesty, as published in the UNH Student Rights, Rules, and Responsibilities Handbook (<http://unh.edu/vpsas/handbook/academic-honesty>). By turning in any piece of work in this course, you declare that you have read and understand the policy, and that you did not engage in any form of academic dishonesty as defined in the Handbook.

Plagiarism can take many forms, such as: submitting someone else's work - in whole or in part - as your own; collaborating on answers for individual assignments, or allowing your own work to be used by another student; copying information from a web site or other text without proper documentation; buying a pre-written paper or lab report.

Cheating is mainly concerned with copying on exams or in lab, bringing crib notes into an exam or referring to notes or the textbook or any other source such as a programmable calculator,

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tablet, or cell phone during an exam. All electronic devices must be turned off and put away for the duration of the exam. Ear buds are not permitted.

Any instances of cheating or plagiarism will result in consequences that can range from a failing grade on the assignment for all students involved to dismissal from the University, as defined in the UNH Student Rights, Rules, and Responsibilities Handbook.

UNHM Biology & Biotechnology Facebook Page: At the suggestion of students this page was created for students and alumni to stay connected to what is going on on-campus and opportunities for you to enhance your career and build your professional network. This is professional use of social media and all are expected to become a member. Join right away! <https://www.facebook.com/groups/213917125371339/>

Grading:	3 lecture exams @ 100 points each.....	300
	Best 10 of 13 weekly quizzes <u>taken</u> @ 10 points each.....	100
	Laboratory	<u>125</u>
	Total Points Available.....	525

LECTURE SCHEDULE

Dates	Lecture # & Topic	Chapter #
August 30	1. Introduction; What is Biology? And scientific method	1
September 6	2. The Chemical Context of Life	2
September 13	3. The Molecular Diversity of Life	3
September 20	4. A Tour of the Cell	4
September 27	5. Membrane Transport and Cell Signaling	5
October 4	<i>Exam 1</i>	
October 11	6. The Cell Cycle and Meiosis	9-10
October 18	7. Inheritance, Genes and Chromosomes	11-12
October 25	8. The Molecular Basis of Inheritance	13
November 1	9. Gene Expression: From Gene to Protein	14

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November 8	Exam 2	
November 15	10. Gene Regulation, Stem Cells, and Viruses	15-17
November 22	11. Genomes, Recombinant DNA and Biotechnology	18
November 29	12. Introduction to Metabolism and Cellular respiration	6-7
December 6	13. Cellular Respiration, Photosynthesis	7, 8
December 13	Final exam	

BIOL 413 Laboratory

Fall 2016

Contact Information:

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Rm 550B

Feel free to contact me at any time. Email works best, I will attempt to respond to you as soon as possible.

Lab Schedule:

Module 1: Scientific Method

Aug 31, Sep 1, 2	Safety, Introduction to Brine Shrimp and Scientific Method
Sep 7, 8, 9	Experimental Design
Sep 14, 15, 16	Run Experiment, Data Collection, Statistical Analysis
Sep 21, 22, 23	Communicating in Science; Anatomy of a Journal Article, Conventions, Presentations **Quiz 1 brine shrimp
Sep 28, 29, 30	Student Presentations

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Module 2: Cells

Oct 5, 6, 7 Mitosis, Meiosis and Cell Division

Module 3: Molecular Genetics

Oct 12, 13, 14 Crime Scene Evidence Collecting & Blood Typing
 ****Quiz 2 Mitosis/Meiosis**

Oct 19, 20, 21 Crime Scene ELISA Test
 ****Quiz 3 Blood Typing**

Oct 26, 27, 28 Crime Scene Restriction Fragment Analysis
 **** Quiz 4 ELISA**

Nov 2, 3, 4
report Crime Scene Final Analysis & Communicating in Science – Forensic report

Nov 9, 10, 11 **Veterans Day No Labs**

Nov 16, 17, 18 Student Presentations
 **** Quiz 5 RFLP**

Nov 23, 24, 25 **Thanksgiving No Labs**

Module 4: Cellular Respiration and Metabolism

Nov 30, Dec 1, 2 Metabolism

Dec 7, 8, 9 **Lab Exam**

Grading:

Lab grade will be a total of 125pts made up of the following:

Presentation/Lab Report from Module 1	20pts
Presentation/Lab Report from module 3	20pts
Lab Report from Module 2	10pts
Lab Report from module 4	10pts
Lab Quizzes	25pts
Lab Exam	20pts
Attendance	20pts
Total	125pts

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Attendance:

Attendance in lab is very important as each lab of a module builds upon another. If you miss a lab you will be responsible for gathering the information needed to come to the next lab ready to perform the lab of that week. Also we will be working as groups during this lab and it is not fair to the others in your group to miss lab as that will leave the other members of the group more work to do. In light of these facts **5 pts will be deducted from your lab grade for each missed lab**. Any more than 2 missed labs will result in possible failure of the lab.

Canvas:

All of the materials for this lab are found in your lab section of Canvas (MyCourses.UNH.edu). While there may be times where you will be given hard copies of materials you will need for lab there will also be times where you will need to print out and bring the materials to lab. It is also a good way for you to review any videos or interactive programs that we may use in lab.

Electronic Devices:

You are welcome to use electronic devices (cell phones, tablets, laptops) in lab for “lab related” purposes. If you however need non-lab related use of these devices (*i.e.*, to take or place a phone call) please be respectful of your labmates and go outside the lab for such activities.

Lab Notebook, Manual, and Coat:

You will need a notebook for lab to record your experimental progress and data in. There is no lab manual that needs to be purchased for this lab. A lab coat is required and must be purchased.

To do well in this lab:

- Attend all labs! If you miss a lab you will miss an experience that cannot be replicated by simply reading the lab.
- Be prepared when coming to lab. Most material is available in advance of the lab meeting time on the Blackboard site.
- Make notes and sketches during lab. Ask questions, ask questions, ask questions!
- Fill out lab reports completely, and answer questions with more than a word or two.
- Study for the practical exam *every week* by reviewing all handouts, lab reports, notes & sketches. If you wait till just before the practical it will be too much information.